INTRODUCTION TO THE FUNDAMENTALS OF THE INTERNET

Prepared by:

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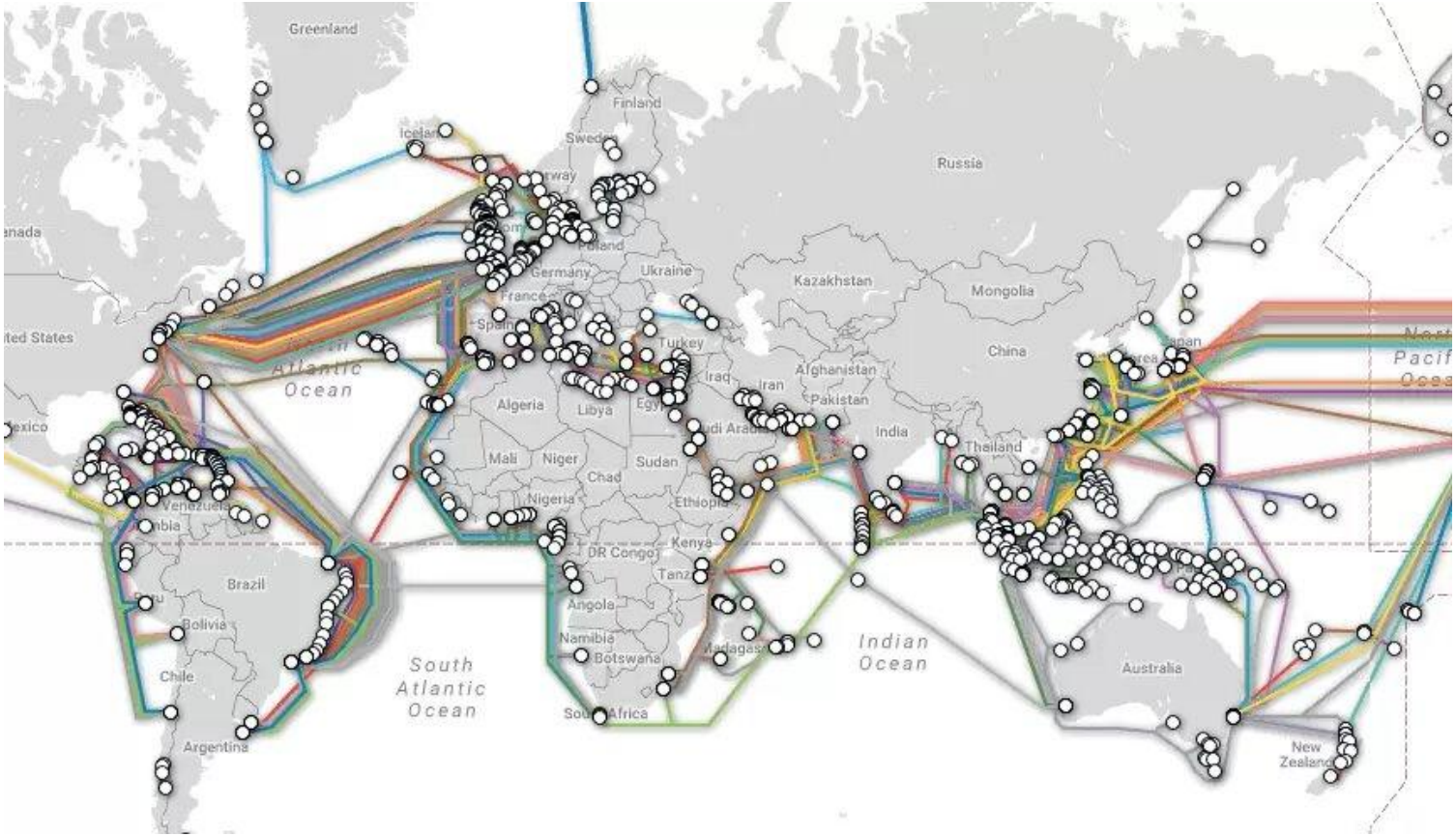
What is internet?

Internet

With the evolution of operating systems and programming languages came the need to connect computers and exchange information. This led to the internet, which consists of fiber optic cable connections spanning the globe.

The Internet is an abbreviation of internetworking, and it was created in the 1970s to refer to the physical cables and computers that are connected to each other.

Reading


Greenland
Finland
Russia
Kazakhstan
Mongolia
China
Germany
Ukraine
Poland
France
Spain
Turkey
Iraq
Iran
Afghanistan
Pakistan
India
Thailand
Algeria
Libya
Egypt
Mali
Niger
Chad
Sudan
Nigeria
Ethiopia
Kenya
DR Congo
Tanzania
Angola
Namibia
Botswana
Madagascar
South Africa
Brazil
Bolivia
Chile
Argentina
Australia
New Zealand
Atlantic Ocean
South Atlantic Ocean
Indian Ocean

Font

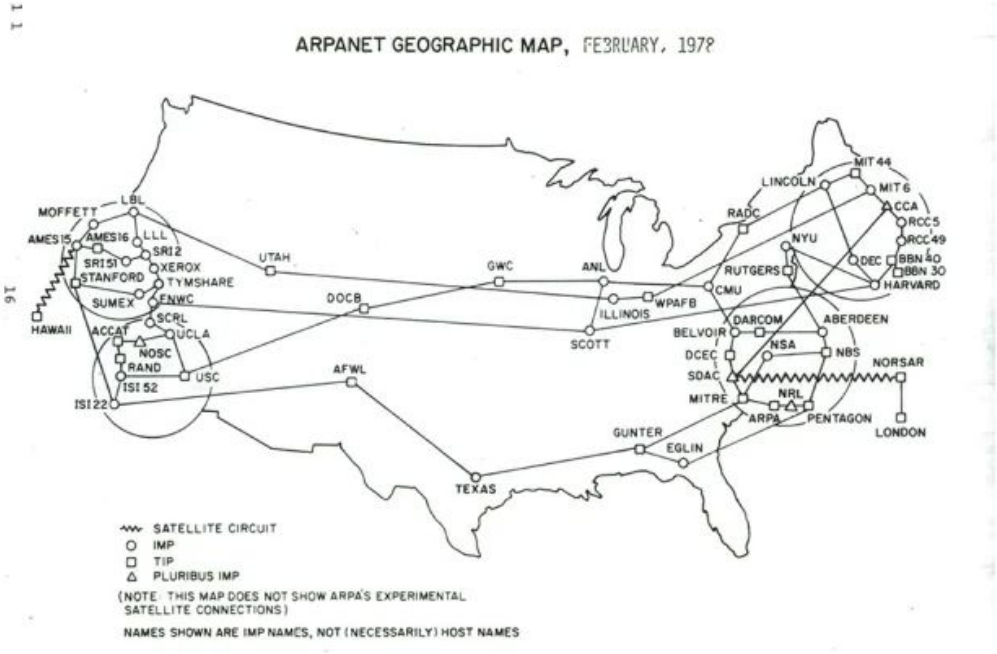
How it all started?

ARPANET

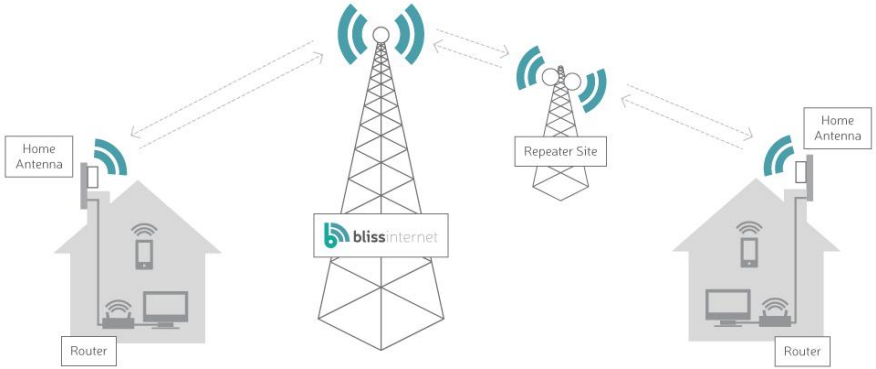
ARPANET (Advanced Research Project Agency NETwork) was an academic research project funded by the ARPA (Advanced Research Projects Agency) of the United States Department of Defense in the 1970s.

Its goal was to develop a way to connect certain universities. It originated in the context of the Cold War (all connectivity in the United States was centralized, and if that connection node were destroyed, the entire country would be left without communication).

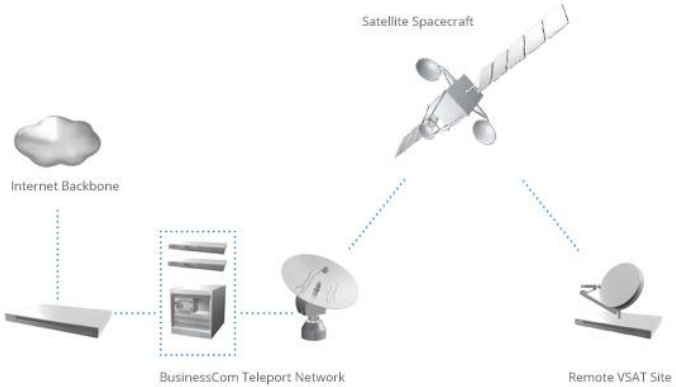
ARPANET



Geographic map of the USA, with all ARPANET stations



Internet wireless



Internet Satellite

What are Internet protocols?

Protocols

Internet protocols are a set of rules and standards that enable communication and data transfer between devices on a network.

- **IP (Internet Protocol)**
- **TCP (Transmission Control Protocol)**
- **FTP (File Transfer Protocol)**
- **HTTP/HTTPS (HyperText Transfer Protocol)(...Secure)**
- **SMTP (Simple Mail Transfer Protocol)**

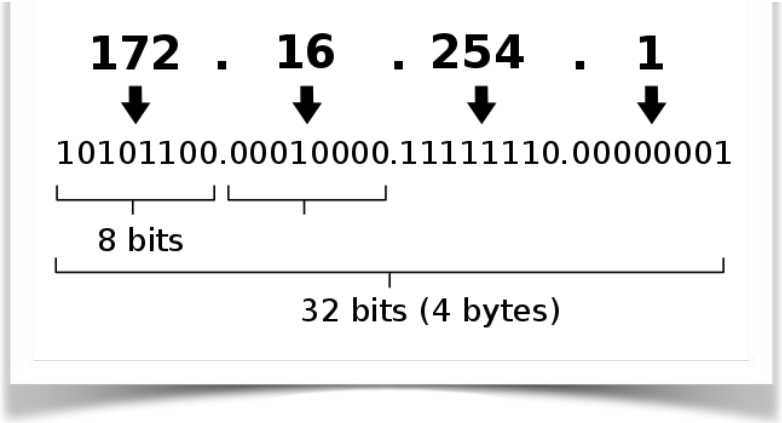
IP protocol

IPv4

IPv6

172.16.256.1

IPv4


172 . 16 . 254 . 1
10101100.00010000.11111110.00000001
8 bits
32 bits (4 bytes)

IPv4

3002:0bd6:0000:0000:0000:ee00:0033:6778

IPv6

IPv4 vs IPv6

Feature	IPv4	IPv6
Address length	32 bits	128 bits
Address representation	4 decimal numbers from 0-255 separated by periods	8 groups of 4 hexadecimal digits separated by colons
Address types	unicast, multicast, broadcast	unicast, multicast, anycast
Packet header	20 bytes long	40 bytes long, but simpler than IPv4 header
Configuration	manual, DHCP	manual, DHCP, auto-configuration
IPSec support	optional	built-in

IPv4 vs IPv6

IP addresses are a non-renewable resource.

Since 1996, the total exhaustion of IPv4 addresses has been announced. Despite the effective use of NAT, by 2015 the entities delegated by the IANA (Internet Assigned Numbers Authority) began announcing the depletion of their assigned ranges. It was expected that at some point in 2019 there would be no new addresses available in any region.

IPv6 has a 128-bit address space, much larger than IPv4's 32-bit space. It has a lighter header and was designed with security standards and traffic prioritization in mind.

The implementation of IPv6 has been slow. According to the Internet Society, in 2018 only 25% of networks connected to the Internet were advertising IPv6 addresses. Only 49 countries had at least 5% of their traffic over IPv6, and of these, only 24 reached 15%.

APPLICATION	HTTP / FTP
TRANSPORT	TCP, UDP
NETWORK	IP, ROUTERS
DATA LINK	ETHERNET, SWITCHES
PHYSICAL	CABLES

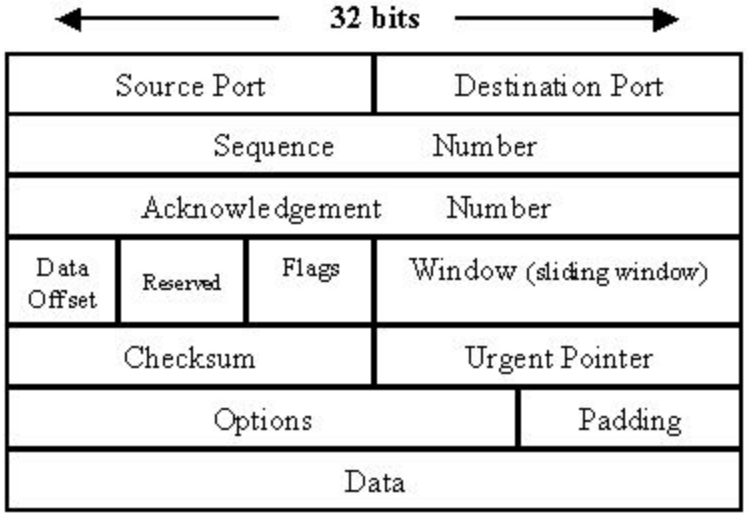
TCP/IP

Protocolo TCP/IP

This protocol is the way in which information is generated and sent over the Internet, through headers that directly contain the address of the intended recipient, which is known as the IP address.

This address is included in the header. When the information reaches an Internet router, the router reads the header and identifies which IP address the information corresponds to. It divides the information into different packets, and across the Internet each packet follows the fastest route to reach the IP address. Automatically, all the information is reassembled, and the IP address—meaning that computer—is the one that receives the information.

TCP/IP protocol structure



APPLICATION	HTTP / FTP
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HTTP/HTTPS

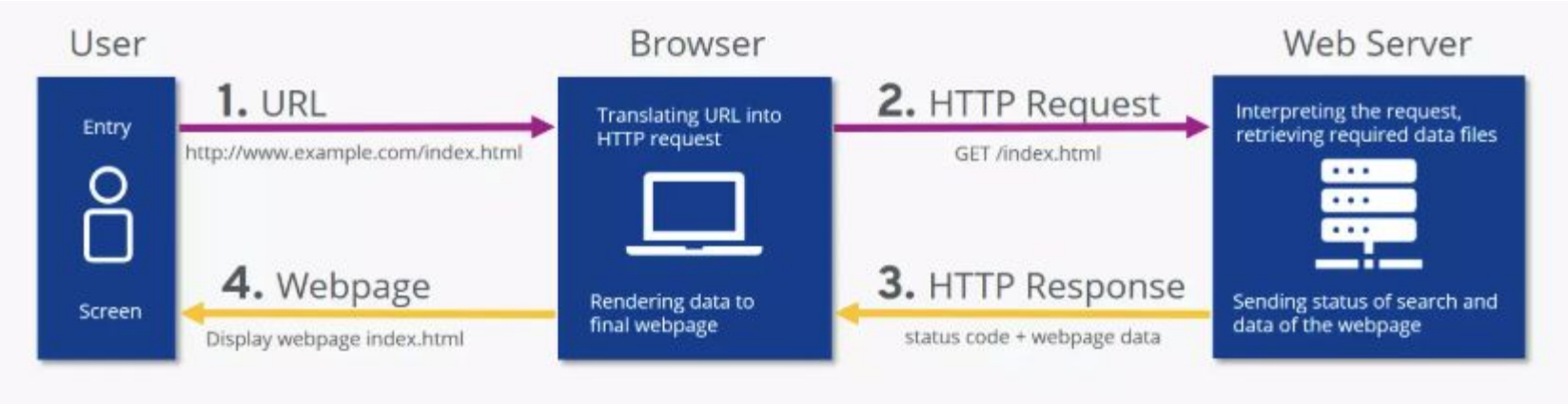
HTTP/HTTPS Protocol

HTTP (HyperText Transfer Protocol): this protocol is based on communication rules. It originated from communication protocols between multiple machines, then continued to evolve along with TCP/IP to identify the IP addresses of computers, which made connections more secure.

HTTP is a protocol oriented toward the Web. It allows access to all information on the Web. It works by sending information that becomes the body of the packet, along with the exact IP address of the final node.

HTTPS is the implementation of security improvements to the HTTP protocol, making the client-server connection secure through encryption algorithms.

HTTP/HTTPS Protocol



METHOD	ACTION
GET	Get data
POST	Send data
PUT	Create / Update data
DELETE	Delete specific data

HTTP methods

HTTP methods

Endpoint	Method	Body	Response	Status
/products	GET	N/A	[{...},{...},...]	200
/products/:id	GET	N/A	{...}	200
/products	POST	{...}	{...}	201
/products/:id	PUT/PATCH	{...}	{...}	201
/products/:id	DELETE	N/A	✅	201

404

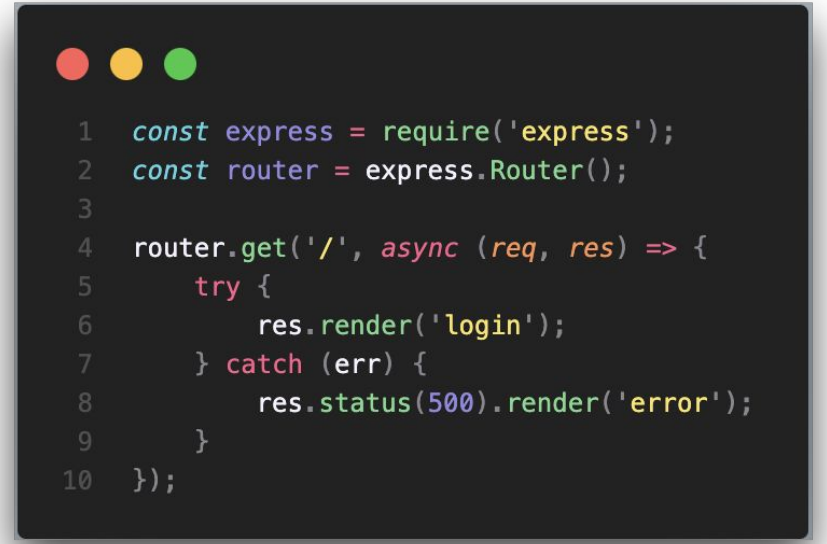
Response codes

Response codes

HTTP response codes are a set of status codes issued by a server in response to an HTTP request from a client. These codes indicate the result of the request and help developers and users understand what has occurred. They are classified into five categories:

- 100 a 199 Informatives
- 200 a 299 Correct
- 300 a 399 Redirection
- 400 a 499 Client side Error
- 500 a 499 Server side Error

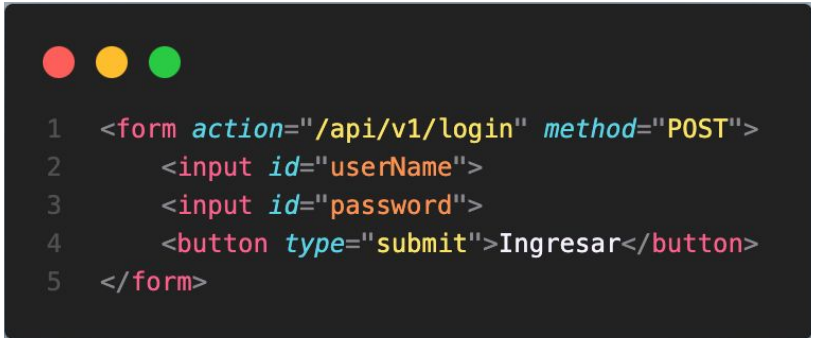
Reading



```
const express = require('express');
const router = express.Router();

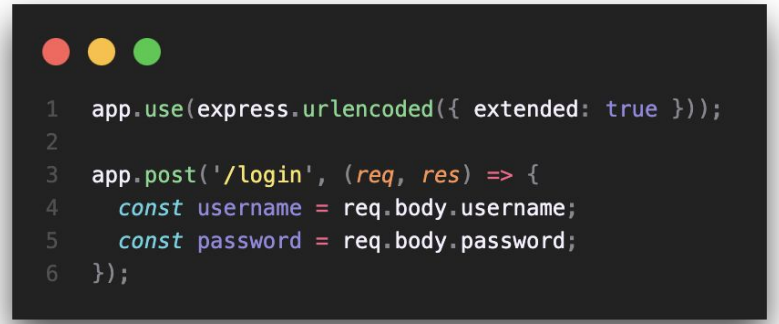
router.get('/', async (req, res) => {
    try {
        res.render('login');
    } catch (err) {
        res.status(500).render('error');
    }
});
```

GET



```html
<form action="/api/v1/login" method="POST">
    <input id="userName">
    <input id="password">
    <button type="submit">Ingresar</button>
</form>
```

POST HTML



```
app.use(express.urlencoded({ extended: true }));

app.post('/login', (req, res) => {
  const username = req.body.username;
  const password = req.body.password;
});
```

POST

DNS	IP
www.google.com	**142.250.78.164**
www.facebook.com	**163.70.152.35**
www.youtube.com	**142.251.132.142**
www.amazon.com	**13.35.138.61**

DNS


~
nslookup www.facebook.com
Server: 192.168.101.1
Address: 192.168.101.1#53
Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name: star-mini.c10r.facebook.com
Address: 163.70.152.35

DNS

DNS

Because addresses are represented by numbers, these numbers are interpreted by machines and are not suitable for humans, so it was necessary to invent a way to translate numbers into a more user-friendly form.

The service that performs this translation is called DNS. With DNS, it is possible to point to a website, and a request made using a domain name will be translated into a number that ultimately reaches the site where the physical machine hosting the website's content is located.

These translations have classifications known as records; that is, for each number a record is added that indicates what it represents and what information the record contains.

Reading











HOSTING

Hosting

Web hosting is a service that allows individuals and organizations to make their websites accessible on the Internet. Web hosting providers offer space on physical or virtual servers where the files and data necessary for a website to function are stored. In addition to storage, these services usually include technical support, security, and management tools such as control panels. There are different types of hosting, such as shared, VPS, dedicated, and cloud hosting, each with distinct features and resource levels to meet the specific needs of each web project.

END